

Nicholas Bondar

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Education

Harvard University, S.B. in Mechanical Engineering, GPA 3.72

Cambridge, MA

Sept 2025 – May 2029

- **In progress:** Engineering Thermodynamics, Mechanical Systems. **Spring 2027:** Fluid Mechanics, Solid Mechanics. **Completed:** Computer Aided Machine Design, Circuits and Digital Systems, Mechanics and Special Relativity

Experience

Harvard Undergraduate Aerospace Collective – Rocketry (HUAC-R), Chief Engineer

Cambridge, MA

Sept 2025 – present

- Designed FUD-1, a 250 lbf regeneratively cooled liquid bipropellant rocket engine (nitrous oxide / 90 percent ethanol, 200 psia chamber), owning combustion, injector, cooling, structural, and feed system design through release to print
- Sized the chamber and nozzle from NASA chemical equilibrium data in a parametric spreadsheet and verified against RPA; traced a C^* and Isp discrepancy to finite- versus infinite-area-combustor modeling and designed to 180.9 s delivered performance
- Reduced peak gas-side wall temperature from roughly 1400 K to 636 K by selecting O/F 3.0, a 90/10 ethanol-water blend, 2 percent film cooling, and 38 axial regenerative channels sized to hold turbulent coolant flow
- Ran hoop and thermal stress analysis on the chamber walls across four load cases, sizing wall thickness to safety factors of 2.7 to 8.0 and identifying low-cycle fatigue as the life-limiting mode
- Sized a 24-orifice oxidizer-centered pintle injector and designed the feed system, thrust stand, and instrumentation, siting transducers so regenerative pressure drop is measured by difference and can trigger an abort; co-authored the standard operating procedure submitted for Harvard EHS approval of cold flow and static fire testing
- Ran an external design review with a former Firefly Aerospace fluids lead and a Relativity Space contractor before committing to an irreversible print, then revised the design accordingly, filleting the internal cooling channel corners where thermal stress concentrates

Harvard Microrobotics Laboratory, Undergraduate Researcher

Cambridge, MA

Feb 2026 – present

- Developed a fabrication process for RoboBee insect-scale composite wings on a \$6,000 laser cutter in place of \$500,000-class equipment, holding 20-50 micron feature tolerances
- Characterized cut recipes across carbon fiber, Mylar, Kapton, and FR4; measured an 85 micron beam kerf and offset cut profiles to hold 95 micron wing spars; traced adhesive damage to substrate conduction and back-reflection and eliminated it with an FR4 interposer, cutting required laser passes from 200 to 160 and enabling clean release of complete wings
- Took a wildlife-tracking drone, a hexacopter carrying a VHF receiver used to locate tagged pangolins in the field, from never powered on to first flight; raised tag detection range 43x from 10 m to 429 m by identifying receiver noise figure as the limiting term and adding a low-noise amplifier

Projects

Two-Wheeled Self-Balancing Robot

2026

- Built the full sensing-to-actuation loop on an ESP32-S3: MPU6050 IMU, complementary filter for angle estimation, and a 200 Hz PID controller, achieving stable upright balance and recovery from disturbances
- Diagnosed a motor driver burnout as insufficient stall-current margin, and a failure to recover from a raised center of mass as a gearing rather than tuning limit; exhibited at the Harvard SEAS Design Fair

Skills

CAD: SolidWorks, Onshape, Fusion 360, AutoCAD, Fusion 360 CAM, GD&T, tolerance and fit analysis

Analysis: SimScale (FEA), NASA CEA, Rocket Propulsion Analysis (RPA), thermal, structural, and fluid hand analysis

Manufacturing: Manual lathe and mill, CNC programming, laser cutting and LightBurn, additive manufacturing, DFM

Programming: Python, C, C++, SQL, JavaScript, HTML/CSS, Arduino and ESP32 embedded development, LaTeX